LESSON 1: PLANT TISSUES

Meristems, simple tissues, and the three tissue systems

GENERAL BOTANY | MIDTERM: PLANT TISSUES, ROOT, STEM, LEAF

How are plant tissues sorted?

Meristematic tissues (young cells that keep dividing) vs **permanent tissues** (mature, specialized, usually stop dividing). Permanent tissues are **simple** (one cell type) or **complex** (several cell types working together). "Simple" and "complex" are about the **number of cell types**, not difficulty.

MERISTEMATIC TISSUES

- **Meristem** – from Greek **meristos, "divided"**; unspecialized cells that **divide indefinitely** (like plant stem cells)
  - Usually found at the **apex (tips) of roots and shoots**
- Characteristics:
  - **Living, thin-walled**
  - **Few, small vacuoles**
  - **Dense protoplasm, conspicuous (large) nuclei**
  - **Spherical, oval or polygonal**, closely packed, little or no intercellular space
  - **Do not store reserve food**; always in active metabolism

BY ORIGIN

- **Promeristem** – earliest, **embryonic** meristem at the extreme tips; made of **initials** (keep dividing); produces the **primary meristems**
- **Primary meristem** – from the promeristem, just below it; forms **primary tissues → increase in length**
- **Secondary meristem** – from **permanent tissues that regain the ability to divide (dedifferentiation)**; → **increase in thickness**
  - **Vascular cambium** – between xylem and phloem; makes **secondary xylem (inside, wood)** and **secondary phloem (outside, inner bark)**; growth rings
  - **Cork cambium** – makes the **periderm** (secondary dermal tissue): **cork outside**, **phelloderm inside**
- Sequence: **Promeristem → Primary meristems → Primary permanent tissues**

BY POSITION

- **Apical – tips of roots and shoots → primary growth, increases length**; produces the 3 primary meristems
  - **Protoderm → epidermis**; **Ground meristem → ground tissue** (cortex, pith); **Procambium → primary xylem and phloem**
- **Intercalary – base of leaves, nodes, internodes** ("inserted between") → **longitudinal growth**; lets **grasses regrow after mowing or grazing** (bamboo, sugarcane, rice, corn)
- **Lateral – sides** of stems and roots → **girth / width (secondary growth)**; = **vascular cambium + cork cambium**; secondary tissues for **support and conduction** (and protection)

Note: Memory tip: Apical = **A**t the tip (taller, longer roots). **I**ntercalary = **I**n between (grass regrows after mowing). **L**ateral = **L**arger girth (wider trunk). If a plant **gets taller but never gets thicker**, suspect the **lateral meristem**.

SIMPLE PERMANENT TISSUES (ONE CELL TYPE)

PARENCHYMA

- **Most abundant** plant cells; in all roots, stems, leaves, fruits; **living**, **thin primary wall**, **very large vacuoles**
- Space-fillers; **photosynthesis**, **storage** of food and water, **metabolism** (respiration, protein synthesis), **wound healing and regeneration**

- Injury → parenchyma regains division → **callus** covers the wound; can regenerate a whole plant (**totipotency**)
- **Chlorenchyma** – parenchyma with **chloroplasts** → **photosynthesis** (leaf mesophyll, green stems)
- **Aerenchyma** – large **air spaces** → **buoyancy** (floating) and **oxygen** to submerged parts (water lily, lotus, rice)

COLLENCHYMA

- Greek **kolla, "glue"**; **living** at maturity, **elongated**
- **Thicker, uneven** primary walls (thickest at the **corners**); cellulose and pectin, **not lignified**
- Below the **epidermis** of young stems, **petioles**, leaf veins
- Main function: **flexible support** for **young, growing** parts (celery "strings"); needs **turgor** too

SCLERENCHYMA

- Greek **skleros, "hard"**; thick **secondary wall** with **lignin**; **mostly dead** at maturity
- **Rigid support and protection** in parts that have **stopped growing**
- **Fibers** – long, in groups/bundles; let stems **move in wind without snapping** (flax, hemp, jute)
- **Sclereids (stone cells)** – cubical or spherical; make structures **rock-hard** (nut shells, seed coats, **gritty pears**)

Note: Slide says lignin makes cells "flexible as well as strong"; the lecture notes correct this: lignin mainly adds **rigidity and hardness**. **Collenchyma = flexible**, **sclerenchyma = rigid**.

Note: Living or dead? Parenchyma **living**, collenchyma **living**, sclerenchyma **dead**.

COMPLEX TISSUES AND THE 3 TISSUE SYSTEMS

- **Complex tissue** – several cell types with one shared function (e.g. **xylem**, **phloem**)
- Three tissue systems, **continuous** through roots, stems and leaves, all from meristems: **Dermal** (protects), **Vascular** (transports), **Ground** (photosynthesis, storage, support)

A. DERMAL TISSUE SYSTEM

- Greek **derma, "skin"**; outer **protective covering** against **injury and desiccation**
- **Epidermis** – usually **one layer** of closely packed cells (from the **protoderm**)
- **Periderm** – replaces the epidermis in plants with **more than one growing season** (woody stems and roots); mainly **dead cork cells** (**suberin**) → protects from **predators and water loss**; made by the **cork cambium**; gas exchange through **lenticels**
- **Trichomes** – hair-like epidermal outgrowths; **block herbivores** physically and chemically (stinging **nettle, Urtica**: histamine and other irritants)
- **Cuticle** – waxy layer of **cutin** on the epidermis → **prevents water loss**; thick in dry habitats, absent on root hairs
- **Guard cells** – epidermal cells **with chloroplasts** that form **stomata** → **gas exchange** (CO_2 in, O_2 and water vapor out)
  - Guard cells take in water → **turgid** → stoma **opens**; lose water → **closes**
- **Root hairs** – epidermal extensions for **absorption**

B. VASCULAR TISSUE SYSTEM

- **Vascular plants = Tracheophyta**; main function **conduction**, extra function **mechanical support**
- **Sap** – the contents moved by xylem and phloem
- **Xylem** – **water and minerals**, **root → upward** (transpiration pull, cohesion, adhesion); also **support**
  - **Tracheids** – **long, tapered**, **dead**, lignified; water passes through **pits**; the **torus** acts as a **valve** (stops air bubbles)

- **Vessels (trachea)** – **vessel elements**: **wider, shorter, less tapered**, **end to end**, open ends = **perforation plates**; side **pits**; dead
- **Xylem fibers** – dead, lignified → **mechanical support** only
- **Xylem parenchyma** – the **only living** xylem cells → **storage** (starch, oil, **ergastic substances**) and **lateral transport**

- **Phloem** – **food (sugars) and organic nutrients**, from **source to sink** (up or down)
  - **Primary phloem** from the **procambium**; **secondary phloem** from the **vascular cambium**
  - **Sieve-tube elements** – **living** but **no nucleus** at maturity; no lignin; joined end to end into **sieve tubes**; end walls = **sieve plates** (pores)
  - **Companion cells** – **have a nucleus**; supply **proteins and energy** to the sieve tube; linked by **plasmodesmata**; load sugar at the source
  - **Phloem parenchyma** – living, **stores starch**
  - **Phloem fibers (bast)** – long, **support**

Note: Slide lists "sieve tubes and sieve plates" as the two sieve elements; the notes correct this: a **sieve tube** is a column of elements and a **sieve plate** is the porous end wall.

Note: Living vs dead: in xylem only **xylem parenchyma** lives; in phloem the **sieve tubes, companion cells and parenchyma** live (fibers are dead).

C. GROUND TISSUE SYSTEM

- Also called the **fundamental tissue system**; everything **not dermal or vascular**
- **Cortex** – ground tissue **between dermal and vascular** tissue
- **Pith** – ground tissue **inside / at the center** of the vascular tissue
- Mostly **parenchyma**, with some collenchyma and sclerenchyma: photosynthesis, storage, support

Note: Memory tip: **D**ermal **D**efends, **V**ascular **V**ehicles (transport), **G**round does the **G**eneral work. **X**ylem = e**X**its the root going up (water); **Ph**loem = **Ph**ood.

SUMMARY TABLE: MERISTEMS

Meristem	Where	What it does	Example / product
Promeristem	extreme tips; embryo	earliest meristem; makes the primary meristems	made of **initials**
Primary meristem	just below the promeristem	primary tissues → **length**	protoderm, ground meristem, procambium
Secondary meristem	from permanent cells that divide again	secondary tissues → **thickness**	vascular cambium, cork cambium
Apical	**tips** of roots and shoots	**primary growth: length**	makes the 3 primary meristems
Intercalary	**base of leaves, nodes, internodes**	**length between nodes**; regrowth after cutting	grasses, bamboo, sugarcane
Lateral	**sides** of stems and roots	**secondary growth: girth**	**vascular cambium**, **cork cambium**

SUMMARY TABLE: PRIMARY MERISTEMS AND WHAT THEY BECOME

Primary meristem	Becomes
Protoderm	**epidermis** (dermal tissue)
Ground meristem	**ground tissue**: cortex and pith (parenchyma, collenchyma, sclerenchyma)
Procambium	**primary xylem and primary phloem** (vascular tissue)

SUMMARY TABLE: SIMPLE TISSUES

Tissue	Alive?	Wall	Where	Function
Parenchyma	**living**	**thin** primary wall; large vacuole	throughout the plant	**photosynthesis**, **storage**, **wound healing**, regeneration
Chlorenchyma	living	thin; **chloroplasts**	leaf mesophyll, green stems	**photosynthesis**
Aerenchyma	living	thin; **air spaces**	aquatic plants	**buoyancy**, oxygen to submerged parts
Collenchyma	**living**	**unevenly thickened** primary wall	**below the epidermis**, petioles, leaf veins	**flexible support** for young parts
Sclerenchyma	**dead**	thick **lignified** secondary wall	wood, bark, nut shells, seeds	**rigid support**, protection
Fibers	dead	long, lignified	in bundles in stems, bark	stems **bend in wind without snapping**
Sclereids	dead	cubical or spherical, very thick	nut shells, seed coats, pears	**rock-hard**, inflexible

SUMMARY TABLE: TISSUE SYSTEMS

System	Made of	Function
Dermal	**epidermis** (young), **periderm** (older woody parts); trichomes, cuticle, guard cells, root hairs	**protection** from injury and **water loss**; gas exchange (stomata); absorption (root hairs)
Vascular	**xylem** + **phloem**	**conduction** of water, minerals and food; **support**
Ground	mostly **parenchyma** (+ collenchyma, sclerenchyma); **cortex** and **pith**	photosynthesis, **storage**, support

SUMMARY TABLE: XYLEM AND PHLOEM CELLS

Cell	Alive?	Key feature	Job

Cell	Alive?	Key feature	Job
Tracheid	dead	long, **tapered**; water through **pits**; **torus** = valve	**water conduction**, support
Vessel element	dead	**wider, shorter**, end to end; **perforation plates**	**fast water conduction**
Xylem fiber	dead	thread-like, lignified	**support only**
Xylem parenchyma	**living**	the **only living** xylem cell	**storage** (ergastic substances), lateral transport
Sieve-tube element	**living**	**no nucleus** at maturity; sieve plates	**carries food** (sugars)
Companion cell	**living**	**has a nucleus**; linked by **plasmodesmata**	**supplies proteins and energy** to the sieve tube
Phloem parenchyma	living	thin-walled	**stores starch**
Phloem fiber	dead	long (bast fibers)	**support**

——— END OF LESSON ———

LESSON 2: ROOT SYSTEM

Structure, modifications, storage, respiration, microbes

GENERAL BOTANY | MIDTERM: PLANT TISSUES, ROOT, STEM, LEAF

How do you know it is a root?

Roots are **non-green**, have **no nodes, internodes, leaves or buds**, have a **root cap**, and grow **toward gravity and water, away from light**. A stem, even underground, has **nodes, internodes and buds**.

CHARACTERISTICS OF THE ROOT

- **Descending** portion of the plant; **non-green, brownish**
- **Nodes, internodes, leaves and buds absent**
- **No lateral branches** like stems (but it does form **lateral roots** from the **pericycle**, inside)
- Tip covered by a **root cap (calyptra)**
- **Positively hydrotropic** (toward water), **negatively phototropic** (away from light), **positively geotropic** (with gravity)
- Vascular bundles **radial**; xylem **exarch** (xylem and phloem on separate radii)
- **Unicellular root hairs** near the tip → **water absorption**

TWO ROOT SYSTEMS

- **Tap root system** – from the **radicle**; **one deep main root**, thick at the base, tapering; **secondary and tertiary** roots; **common in dicots**; reaches **deep** water
- **Adventitious root system** – **many roots of equal length**; **surface feeders**; arise from the **stem, leaf or hypocotyl** (not the radicle); underground or aerial, green or non-green; **common in monocots** (fibrous: grasses, corn)

Note: Slide: adventitious roots lack root caps and root hairs. Notes: that is only true for some **specialized aerial** roots; ordinary soil adventitious roots usually have them.

PARTS (ZONES) OF A TYPICAL ROOT, TIP UPWARD

- **1. Root cap (calyptra)** – multicellular, **parenchymatous**
  - **Protects** the growing point from **soil friction**
  - **Rich in starch** (amyloplasts/statoliths) → **senses gravity (gravitropism)**
  - **Rich in Golgi bodies** → outer cells secrete **mucilage** that **lubricates** the tip
  - Replaced by **root pockets** in **Eichhornia, Pistia, Lemna** (balance in water); **multiple root caps** in **Pandanus** stilt roots; reduced or absent in **mycorrhizal** roots
- **2. Growing point (meristematic zone)** – about **1 mm**; **cell division**; toward the tip → **calyptrogen** (makes the cap); toward the base → **histogen**
- **3. Zone of elongation** – **4–8 mm**; cells **lengthen by vacuolization**; **division stops**; pushes the tip deeper
- **4. Root hair zone (piliferous region)** – **main water and mineral absorption**; root hairs **unicellular and short-lived (ephemeral)**; the zone keeps "moving forward" with the tip
- **5. Zone of maturation** – above the hair zone; the **bulk** of the root; thick walls, **little absorption**; **lateral roots** arise here; **secondary growth**

Note: Memory tip: tip upward: **C**ap, **D**ivide, **E**longate, **H**airs, **M**ature: "**C**an **D**arryl **E**at **H**am **M**ondays?"

FUNCTIONS OF ROOTS

- Primary: **anchorage, absorption, conduction, storage**
- Special functions and examples (slide list):
  - **Food storage** – carrot, radish; **extra support** – banyan, maize

- **Aeration** – mangroves; **reproduction** – dahlia, sweet potato
- **Floating** – Jussiaea; **photosynthesis** – Tinospora; **green roots** – Trapa
- **Nitrogen fixation** – pea (Rhizobium nodules)
- **Food from a host** – **Cuscuta** (haustoria); **water from the air** – **Vanda** (velamen)
- **Climbing** – Pothos; **phosphate absorption** – mycorrhizal roots; flattened "plantless" roots – Podostemum

ROOT MODIFICATIONS

TAP ROOT FOR STORAGE

- Tap root becomes **swollen and fleshy**; food used at **flowering and fruiting**; secondary roots stay thin; stem **reduced, disc-like**, with **radical leaves**
- **Conical** – broad at the top, tapers down (cone) → **carrot**
- **Fusiform** – **spindle**: thick in the **middle**, tapering both ends; hypocotyl is 2/3 of it → **radish**
- **Napiform** – **almost spherical** top, **suddenly thin** below → **turnip** (beetroot)
- **Tuberous (tap)** – swollen, **no definite shape**, single → **Mirabilis** (four-o'clock)

Note: Memory tip: Cone = **C**arrot. **F**usiform = **F**at middle (radish). **N**apiform = **N**ice round top (turnip).

ADVENTITIOUS ROOTS FOR STORAGE AND SUPPORT

- **Tuberous** – fleshy, no definite shape, **single at each node**, starch → **sweet potato**
- **Fasciculated** – **clusters** from the lower nodes → **dahlia** (stores **inulin**)
- **Moniliform** – **beaded** (Portulaca); **Nodulose** – swollen **tips** (mango ginger); **Annulated** – **ring-like** swellings (Psychotria)
- **Stilt** – from the **lower nodes** into the soil, **support** → **sugarcane, maize**
- **Palmate** – like the **palm of a hand** (Orchis)
- Diagram groups: adventitious **mechanical support** = prop, stilt, climbing, buttress; **vital functions** = epiphytic, foliar, sucking, photosynthetic

Note: Sweet potato = root (no eyes or buds); **potato = stem tuber** (has eyes = buds at nodes).

ROOTS FOR RESPIRATION (MANGROVES)

- Mangroves = **halophytes** in **salty, waterlogged, oxygen-poor** mud: hard to **anchor, aerate, germinate**
- **Pneumatophores** – **breathing roots**; grow **upward (negatively geotropic)**; **lenticels** at the tips for **gas exchange**; modified **secondary roots** → **Avicennia, Sonneratia**
- **Stilt roots** in **Rhizophora** – support in soft mud
- **Vivipary** – seeds **germinate while still on the parent**

ROOTS FOR REPRODUCTION

- **Adventitious buds** on roots grow into new plants (sweet potato, dahlia); **root tubers** sprout new plants

ROOTS AND MICROBES

- **Rhizosphere** – **soil around the root**; root exudes vitamins and hormones that **feed microbes**
- **Rhizoplane** – the **root surface** where microbes attach
- **PGPR** (slide: PGRR) – **plant growth-promoting rhizobacteria**
- 4 interactions: **mycorrhiza**, **reciprocal symbiosis in nodules** (Rhizobium), **associative symbiosis**, **mycotrophy**
- **Mycorrhiza** – **fungus + root**; **fungal hyphae replace root hairs** (Pinus); roots shallow, no cap or hairs
  - **Ectotrophic** – fungus mostly **outside**, **woolly sheath** → Pinus, Eucalyptus, oak
  - **Endotrophic** – fungus **inside cortex cells**: **vesicles** (store lipids) and **arbuscules** (pass

nutrients); orchids, crops, grasses; **VAM → phosphate absorption**

Note: Memory tip: Ecto = Exterior sheath; **Endo = iNside** the cells.

SUMMARY TABLE: TAP ROOT VS ADVENTITIOUS ROOT

	Tap root	Adventitious root
Origin	**radicle** of the seed	**stem, leaf or hypocotyl** (not the radicle)
Look	**one main root** with side roots	**many roots of equal size**
Depth	**deep**	**surface feeder**
Common in	**dicots** (carrot, mango)	**monocots** (grasses, corn)

SUMMARY TABLE: ZONES OF A TYPICAL ROOT (TIP UPWARD)

Zone	Size	What happens
Root cap (calyptra)	tip	**protects** the tip from soil friction; **starch → senses gravity**; **Golgi → mucilage** lubricates
Meristematic zone	**~1 mm**	**cell division**; calyptrogen (to the cap) and histogen (to the base)
Zone of elongation	**4–8 mm**	cells **lengthen by vacuolization**; division stops; pushes the root down
Root hair zone	temporary	**main absorption** of water and minerals; hairs unicellular, short-lived
Zone of maturation	bulk	thick walls, little absorption; **lateral roots** arise; **secondary growth**

SUMMARY TABLE: STORAGE ROOTS

Type	From	Shape	Example
Conical	tap root	broad at the top, tapers down (cone)	**carrot**
Fusiform	tap root	**spindle**: thick in the middle	**radish**
Napiform	tap root	**almost spherical**, suddenly thin below	**turnip**
Tuberous	tap root	swollen, no definite shape	**Mirabilis**
Tuberous	adventitious	fleshy, no shape, single at each node	**sweet potato**
Fasciculated	adventitious	**clusters**; stores **inulin**	**dahlia**
Moniliform	adventitious	**beaded**	Portulaca
Nodulose	adventitious	swollen **tips**	mango ginger
Annulated	adventitious	**ring-like** swellings	Psychotria

SUMMARY TABLE: OTHER ROOT MODIFICATIONS

Root	What it does	Example
Pneumatophores	grow **upward**; **lenticels** for **gas exchange**	**Avicennia, Sonneratia** (mangroves)
Stilt roots	from **lower nodes**; **extra support**	**sugarcane, maize**, Rhizophora
Prop roots	support heavy branches	**banyan**
Climbing roots	cling to a support	**Pothos**
Sucking roots (haustoria)	take food from a **host**	**Cuscuta**
Aerial (velamen) roots	absorb **moisture from the air**	**Vanda**
Nodules	**nitrogen fixation** (Rhizobium)	**pea** (legumes)
Mycorrhizal roots	fungus **replaces root hairs**; **phosphate** uptake	**Pinus** (ecto), orchids (endo, VAM)

——— END OF LESSON ———

LESSON 3: STEM

External structures, texture, and stem modifications

GENERAL BOTANY | MIDTERM: PLANT TISSUES, ROOT, STEM, LEAF

What makes something a stem?

It has **nodes, internodes and buds** (terminal and axillary), and its leaves or scale leaves sit at the nodes. This is how a **potato, ginger, onion or taro** underground is still a **stem**, not a root.

THE STEM

- The axis that bears **buds and shoots with leaves**, with roots at its base; the "**central axis**"; part of the **shoot system**; **phototropic** (grows toward light)
- First stem comes from the **epicotyl** (embryonic shoot in the seed)
- **Lateral branches** arise in the **axils** of leaves (angle between leaf and stem)
- Characteristics: usually **above ground**; bears **nodes, internodes, leaves, flowers**; **transports** (root → leaves and leaves → roots); **stores** nutrients

EXTERNAL STRUCTURES

- **Bud** – primary growing point; **leaf bud (vegetative)** or **flower bud (reproductive)**
- **Terminal (apical) bud** – at the tip; releases **auxin**, which **suppresses lateral buds** → growth goes **up, not out** (**apical dominance**)
  - **Cut the terminal bud** → lateral buds grow → **bushier, shorter** plant (why pruning causes branching)
- **Lateral (axillary) buds** – in the **leaf axils**; become branches or flowers
- **Bud scales** – **modified leaves** covering and protecting a bud
- **Leaf scar** – mark left when a **leaf falls**
- **Bundle scar** – marks **inside the leaf scar** from the **vascular bundles**; used to **identify woody plants**
- **Lenticels** – **pores for gas exchange** in bark (cannot open and close like stomata)
- **Node** – where **leaves and lateral buds** attach; **Internode** – the part **between two nodes**
- **Bark** – protective outer tissue of woody stems; develops **with age** (all tissues outside the vascular cambium)

TEXTURE OF STEMS

- **Herbaceous** – no permanent woody tissue, short life, **dies after flowering** → **banana** (its "trunk" is a **pseudostem** of leaf sheaths)
- **Suffrutescent** – **half-woody**, woody **at the base** → **eggplant, jute**
- **Woody** – permanent woody tissue (**secondary xylem**) → all **shrubs and trees**

UNDERGROUND STEM MODIFICATIONS

- Protect against bad weather and animals; **store food**; **vegetative propagation**; proven stems by **nodes, internodes, scale leaves, axillary buds, terminal bud**; anatomy like an aerial stem
- **Rhizome** – **thick, horizontal**, stores food; nodes, internodes, scale leaves, buds, **adventitious roots** below → **ginger, turmeric**
- **Tuber** – **swollen end of an underground branch (stolon)**; stores **starch**; **eye = leaf scar + axillary bud** → new plants → **potato**
- **Bulb** – stem reduced to a **disc**; food in **fleshy leaves**; adventitious roots below → **onion, garlic**
  - **Tunicated bulb** – fleshy inner leaves, **dry outer** leaves forming a **tunic** (onion); garlic **cloves** = daughter bulbs from axillary buds
- **Corm** – **swollen, solid, vertical base** of an erect stem; food in the **stem**; old corm **shrivels**, a **new corm** forms each season → **Amorphophallus, Colocasia** (gabi)

Note: Memory tip: Rhizome **R**uns sideways (ginger). **T**uber has **eyes** (potato). **B**ulb = **B**unch of leaves (onion). **C**orm = **C**ompact solid stem (gabi).

SUBAERIAL STEM MODIFICATIONS (VEGETATIVE REPRODUCTION)

- **Runner** – from the **base**, runs **along the soil surface**; long internodes; **roots below, leaves above** at each node → **Oxalis, strawberry (Fragaria), Centella**
- **Offset** – **short, thick** runner (one internode) with a new plant at its **tip**; grows **in all directions** → **Pistia, Eichhornia** (water plants)
- **Stolon** – lateral branch from the **underground stem**, grows **horizontally in the soil**, its tip **comes up** as a new plant → **Colocasia**
- **Sucker** – lateral branch near ground level, travels **underground**, then **turns up** → **Chrysanthemum** (banana, mint)
- Any separated part can **live independently** if injured

Note: Runner vs offset: runner is **long and thin** (land); offset is **short and thick** (water plants). **Stolon vs sucker:** stolon grows **horizontally then emerges at its tip**; sucker travels **underground then turns upward**.

AERIAL STEM MODIFICATIONS

- **Stem tendril** – coiling, for **climbing**; from the **terminal bud** (Cissus quadrangularis) or **axillary bud** (passion fruit)
- **Stem thorn** – **hard, straight, pointed**; may bear leaves, flowers, fruits (Duranta, pomegranate); **checks transpiration** and **protects from grazing** → Carissa, Duranta, Citrus
- **Thorn vs spine vs prickle: thorns = modified stems**, **spines = modified leaves**, **prickles = no vascular tissue** (easily removed)
- **Phylloclade** – **flattened green stem** of **several internodes** acting as a **leaf** → **Opuntia** (its spines = modified leaves of the axillary bud)
- **Cladode** – phylloclade of **one or two internodes**; **cladophyll** – flattened leaf-like stem in the axil of a tiny true leaf → **Asparagus**
- **Bulbil** – **multicellular body for reproduction**; **Agave**: floral buds → bulbils; **Dioscorea**: axillary bud → bulbil

SUMMARY TABLE: STEM PARTS

Part	What it is
Terminal bud	tip bud; releases **auxin** → **suppresses lateral buds** (growth goes up)
Lateral (axillary) bud	in the **leaf axil**; becomes a branch or flower
Bud scale	**modified leaf** protecting a bud
Leaf scar	mark left when a **leaf falls**
Bundle scar	marks **inside the leaf scar** from vascular tissue; **identifies woody plants**
Lenticel	**pore for gas exchange** in bark
Node / internode	where leaves and buds attach / the part **between** two nodes
Bark	protective outer tissue of woody stems; develops **with age**

SUMMARY TABLE: UNDERGROUND STEMS

Type	Description	Food stored in	Example
Rhizome	thick, **horizontal**; nodes, scale leaves, buds, adventitious roots	the stem	**ginger, turmeric**
Tuber	**swollen end of an underground branch**; **eyes** = leaf scar + bud	the stem (**starch**)	**potato**
Bulb	stem reduced to a **disc**; **tunicated** in onion	**fleshy leaves**	**onion, garlic**
Corm	**swollen solid vertical base** of an erect stem; new corm each season	the stem	**Colocasia, Amorphophallus**

SUMMARY TABLE: SUBAERIAL AND AERIAL STEMS

Type	Description	Example
Runner	from the base, **runs along the soil**; roots below, leaves above at nodes	**Oxalis, strawberry, Centella**
Offset	**short, thick** runner; new plant at its tip; all directions	**Pistia, Eichhornia**
Stolon	from the **underground stem**, grows horizontally, **tip emerges** as a new plant	**Colocasia**
Sucker	near ground level, **travels underground**, then **turns up**	**Chrysanthemum**
Stem tendril	coils for **climbing**; from terminal or axillary bud	**Cissus**, passion fruit
Stem thorn	**hard, pointed modified stem**; protection, less transpiration	**Duranta, Carissa, Citrus**
Phylloclade	**flat green stem** of several internodes acting as a leaf	**Opuntia**
Cladode / cladophyll	phylloclade of 1–2 internodes / leaf-like stem in a tiny leaf's axil	**Asparagus**
Bulbil	**bud** that drops off and grows into a new plant	**Agave** (floral bud), **Dioscorea** (axillary bud)

——— END OF LESSON ———

LESSON 4: LEAF

Parts, venation, kinds, shapes, phyllotaxy, modifications

GENERAL BOTANY | MIDTERM: PLANT TISSUES, ROOT, STEM, LEAF

Why study all the leaf terms?

Leaves are the **main vegetative organs** for **photosynthesis, transpiration and gas exchange**. Their parts, venation, arrangement and modifications are how plants are **described and identified**, and each modification solves a problem (climbing, water loss, defense, food capture).

THE LEAF

- **Thin, flat lateral outgrowths** from **stem nodes**
- Develop from **leaf primordia** (outer meristem tissue) → **exogenous** origin
- Each leaf has an **axillary bud** in its axil
- **Heterophylly** – **different kinds of leaves on the same plant**; common in **hydrophytes** (**Limnophila heterophylla**: submerged leaves **dissected**, aerial leaves **entire**); also **coriander**, Eucalyptus, jack
- Size: from minute scales to **palm leaves up to 40 feet**

PARTS OF A DICOT LEAF

- **Leaf base (hypopodium)** – attaches to the stem; the upper angle is the **axil**
  - **Pulvinus** – **swollen leaf base** (legumes) → **pulvinate** leaves (**Mimosa pudica**, mango); causes **sleep movements (nyctinasty)** by turgor changes
  - **Sheathing leaf base** – in **monocots** (banana, grasses); wraps the stem; banana's overlapping sheaths = **pseudostem**
  - **Stipules** – pair of leafy outgrowths at the base; **protect buds**; **stipulate** (Hibiscus, Ixora) vs **exstipulate** (Mangifera); enclose the bud in **Ficus**
- **Leaf stalk (mesopodium)** – **petiole** in a simple leaf, **rachis** in a compound leaf; no stalk = **sessile** (Calotropis, Zinnia); with stalk = **petiolate** (Hibiscus, Ficus); **winged** petiole (Citrus)
- **Lamina / leaf blade (epipodium)** – green, thin, expanded; **photosynthesis, respiration, transpiration**; tip = **apex**, edge = **margin**; **midrib** → **veins** → **veinlets** (framework and transport)

Note: Memory tip: Hypopodium = **low**est (base), **Meso**podium = **middle** (stalk), **Epi**podium = **on top** (blade).

MONOCOT LEAF

- Same 3 parts, but **sheathing leaf base**; **no petiole in grasses** (rachis in some palms)
- **Ligules** – small outgrowths at the base-blade junction (grasses)
- Lamina **isobilateral** (both sides alike); **parallel venation**

VENATION

- **Reticulate** – veins form a **network**; typical of **dicots**
- **Parallel** – veins parallel; typical of **monocots**; **horizontal** (banana) or **longitudinal** (bamboo, grasses, lilies)
- Exceptions: **dicots with parallel** venation – Calophyllum, Alstonia, Eryngium; **monocots with reticulate** – aroids, Dioscorea (yam)

SIMPLE AND COMPOUND LEAVES

- **Simple** – **one lamina**; incisions **do not reach the midrib** (Hibiscus)
- **Compound** – lamina divided into **leaflets (pinnae)** on a **rachis**; leaflets have **no buds or stipules** in their axils
- **Compound leaf vs branch:** compound leaf has **no nodes/internodes, no axillary buds, no**

terminal bud, no flowers or fruits; a branch has all of these

PINNATELY COMPOUND (LEAFLETS ALONG BOTH SIDES OF THE RACHIS, LIKE A FEATHER) – NEEM, MORINGA

- **Unipinnate** – **paripinnate** (**even**, ends in a pair: tamarind) or **imparipinnate** (**odd**, ends in one leaflet: rose, neem)
- **Bipinnate** – pinnate **twice** via **rachillae** (Acacia, Mimosa pudica)
- **Tripinnate** – pinnate **thrice** (Moringa)
- **Decompound** – **more than thrice** (coriander)

PALMATELY COMPOUND (LEAFLETS RADIATE FROM ONE POINT, LIKE FINGERS)

- **Unifoliate** – 1 (Citrus); **Bifoliate** – 2 (Bignonia); **Trifoliate** – 3 (rubber); **Quadrifoliate** – 4 (Marsilea); **Multifoliate (digitate)** – more than 4 (Bombax, tapioca)

Note: Memory tip: Pari = **pair** (even). **Impari** = **impair** / odd. Citrus looks simple but is **unifoliate compound** (look for the joint).

LAMINA SHAPE, MARGIN, APEX, SURFACE

- **Shapes: linear** (grass), **lanceolate** (lance, broad near the base: Nerium), **oblanceolate** (broad near the apex), **elliptical** (guava), **oblong** (plantain), **ovate** (egg, broad at base: banyan), **obovate** (broad near apex), **orbicular** (lotus), **cordate** (heart), **obcordate** (inverted heart: Oxalis leaflets), **reniform** (kidney: Centella), **cuneate** (wedge), **deltoid**, **sagittate** (arrowhead: Sagittaria), **falcate** (sickle: Eucalyptus), **oblique** (unequal halves: Begonia), **lyrate** (mustard), **acicular** (needle: Allium), **lorate** (strap)
- **"Ob-" = the inverted form** (broad part at the tip)
- **Margins: entire** (smooth: mango), **serrate** (saw teeth pointing **up/forward**), **dentate** (teeth pointing **outward**: Hibiscus), **crenate** (rounded teeth), **undulate** (wavy: Polyalthia), **spiny** (pineapple, Aloe), **ciliate** (hairs), **fid** (cut halfway: castor, papaya), **sect** (cut almost to the midrib: tapioca)
- **Apex: acute** (sharp), **acuminate** (long tail: Ficus religiosa), **obtuse** (blunt: banyan), **cuspidate** (rigid spiny point: pineapple), **mucronate** (round with a short sharp point), **retuse** (shallow notch), **emarginate** (deep notch: Bauhinia)
- **Surface: glabrous** (smooth, shiny), **pubescent** (short soft hairs: Hibiscus), **villous** (long soft hairs), **hispid** (stiff hairs), **scabrous** (rough), **tomentose** (matted hairs), **glandular** (sticky), **gland-dotted** (oil spots: Citrus), **strigose**, **sericeous** (silky)

PHYLLOTAXY (ARRANGEMENT OF LEAVES)

- Purpose: give leaves **enough light**
- **Prefoliation** – leaves in the bud; **ptyxis** – how each young leaf is folded; **vernation** – arrangement of leaves in the bud
- **Radical** – leaves clustered on a very short stem at the root top (radish); **Cauline** – leaves on a stem with internodes
- **Alternate** – **one leaf per node**: **spiral** (Hibiscus), **distichous** (2 rows: Annona, grasses), **tristichous** (3 rows: Cyperus), **pentastichous** (5 rows)
- **Opposite** – **a pair per node**: **decussate** (pairs at **right angles**: Ixora, Calotropis) or **superposed** (pairs **directly above** each other: Quisqualis)
- **Whorled** – **3 or more per node** (Nerium/oleander, Alstonia)

MODIFICATIONS OF LEAVES

- **Tendrils** – coiled, **for climbing**, sensitive to touch (Pisum: terminal leaflets; Gloriosa: leaf tips; Lathyrus: whole leaf; Clematis: rachillae)
- **Hooks** – terminal leaflets as **cat-claw hooks** for clinging (Bignonia unguis-cati)
- **Spines** – **xerophytes** (cacti); Opuntia: minute leaves of axillary buds; Ananas, Agave: leaf margins → **defense and less transpiration**

- **Scale leaves** – thin, stalkless, **protect axillary buds**; fleshy storage in **onion**
- **Pitcher** – **lamina** becomes a flask to **trap insects** (Nepenthes, Sarracenia); lid = **operculum** (modified leaf tip); slippery sugary rim; enzymes digest insects
- **Phyllode** – **flattened petiole or rachis** acting as a leaf; **xerophytic**, reduces transpiration (Acacia, Parkinsonia)
- **Phyllode vs phylloclade:** phyllode = **leaf** modification, unbranched, **no nodes**, never succulent; phylloclade = **stem** modification, branched, **nodes present**, succulent
- **Functions of leaves:** photosynthesis; gas exchange and transpiration; storage; vegetative propagation (adventitious buds); defense (spines); climbing (tendrils, hooks); food capture (insectivorous); pollination (bright colors)

SUMMARY TABLE: LEAF PARTS

Part	Key facts
Leaf base (hypopodium)	attaches to the stem; **pulvinus** (swollen, Mimosa: sleep movements); **sheathing** in monocots; **stipules** protect buds
Leaf stalk (mesopodium)	**petiole** (simple leaf) or **rachis** (compound); **sessile** = no stalk
Lamina (epipodium)	blade; **photosynthesis**; apex, margin, **midrib** → **veins** → **veinlets**
Monocot leaf	**sheathing base**, **no petiole in grasses**, **ligules**, **isobilateral**, **parallel venation**

SUMMARY TABLE: COMPOUND LEAVES

Type	Description	Example
Paripinnate	even number of leaflets; ends in a **pair**	**tamarind**
Imparipinnate	odd number; ends in **one** leaflet	**rose, neem**
Bipinnate	pinnate **twice** (rachillae)	**Acacia, Mimosa pudica**
Tripinnate	pinnate **thrice**	**Moringa**
Decompound	pinnate **more than thrice**	**coriander**
Unifoliate	palmate, **1** leaflet	**Citrus**
Bifoliate / trifoliate	**2 / 3** leaflets	Bignonia / **rubber**
Quadrifoliate	**4** leaflets	**Marsilea**
Multifoliate (digitate)	**more than 4**	**Bombax, tapioca**

SUMMARY TABLE: PHYLLOTAXY

Type	Arrangement	Example
Alternate spiral	one leaf per node, spiraling	**Hibiscus**
Alternate distichous	one per node, **2 rows**	**Annona**, grasses
Opposite decussate	pair per node, pairs at **right angles**	**Ixora, Calotropis**
Opposite superposed	pair per node, pairs **directly above** each other	**Quisqualis**
Whorled	**3 or more** per node	**Nerium, Alstonia**
Radical	cluster on a very short stem at the root top	**radish**

SUMMARY TABLE: LEAF MODIFICATIONS

Modification	Part modified	Function	Example
Tendril	leaflets, leaf tip, whole leaf	**climbing**	**Pisum, Gloriosa, Lathyrus**
Hook	terminal leaflets	**clinging**	**Bignonia unguis-cati**
Spine	leaves / leaf margins	**defense**, less transpiration	**Opuntia, Ananas, Agave**
Scale leaf	thin, stalkless leaf	**protects buds**; storage (onion)	**onion**
Pitcher	**lamina** (lid = leaf tip)	**traps insects**	**Nepenthes**, Sarracenia
Phyllode	**petiole or rachis**	photosynthesis, **less transpiration**	**Acacia**, Parkinsonia

——— END OF LESSON ———